

Real-time Object-based Hierarchical 3D Scene Graph Construction for Open-Vocabulary Language-guided Navigation

Abstract—This paper introduces a novel framework for real-time, open-vocabulary, language-guided object navigation in unseen indoor environments. Our approach utilizes an *object-based hierarchical 3D scene graph (3DSG)* for dynamic, structured scene representation. Unlike existing methods that rely on the Segment Anything Model (SAM) and CLIP for segmentation and object reconstruction, we employ a fast, open-set object detection model to construct and continuously update the 3DSG, enabling the agent to identify and locate objects from arbitrary textual prompts in real time. By incrementally building a hierarchical scene graph, the framework accurately captures spatial relationships among objects. Furthermore, through integration with Large Language Models (LLMs), the agent can perform human-like reasoning, prioritizing exploration of relevant areas and efficiently locating target objects in complex, large-scale environments. Experiments on the Habitat-Matterport 3D (HM3D) ObjectNav benchmark demonstrate that our framework outperforms the existing methods in success weighted by path length (SPL), demonstrating its effectiveness as a robust and efficient solution for object navigation tasks. Additionally, evaluations on the ScanNet dataset confirm our framework’s ability to construct accurate 3D scene graphs, further validating its capability to capture spatial relationships.

I. INTRODUCTION

Simultaneously exploring and navigating toward an object specified by arbitrary text in an unseen environment, known as object navigation (ObjectNav), represents a fundamental challenge for intelligent agents. Humans inherently excel at ObjectNav due to their ability to comprehend real-world semantic information among objects and retain a structured memory of spatial layouts. Recent advancements in 3D scene graphs (3DSGs) have demonstrated their potential in enhancing ObjectNav, particularly in robotics, by encoding objects, places, and rooms as nodes and modeling their spatial relationships through graph edges. However, the practical deployment of 3DSGs for real-time navigation remains hindered by two primary challenges. First, existing 3DSG construction methods mostly operate in an offline manner and require extensive computational resources, making them impractical for real-time applications. Enabling online 3DSG generation necessitates a paradigm shift toward efficient, incremental SG construction that continuously updates as the agent explores its environment. Second, previous studies have predominantly focused on closed-set vocabularies, limiting their adaptability to real-world scenarios where novel objects, unspecified during training, must be identified and queried dynamically. Addressing these challenges is inherently complex, as real-time 3DSG generation demands not only computational efficiency but also the capability to accommodate open-set object detection and flexible querying

mechanisms. Object-based approaches offer a promising alternative by structuring SGs around discrete entities rather than relying on exhaustive mesh or point cloud, thereby reducing the computational overhead of full-scene reconstruction while maintaining essential semantic and spatial relationships. This efficiency is particularly critical in dynamic environments, where the ability to update and query the SG in real time can directly impacts the navigation performance.

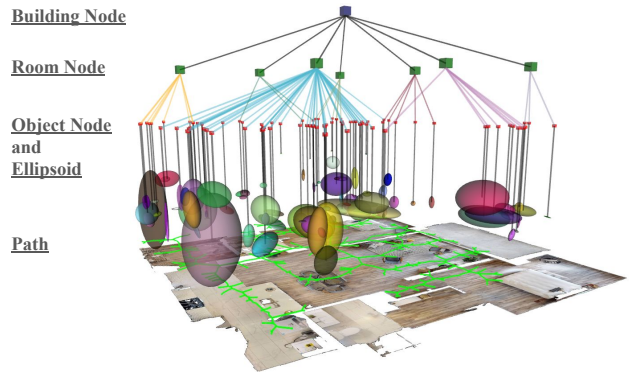


Fig. 1. Visualization of the hierarchical 3D SG generated by our method. The SG organizes the environment into hierarchical levels, where rooms, and objects serve as nodes, while edges denote spatial relationships. Objects are further represented using elliptical approximations, enabling efficient real-time reasoning and semantic understanding for ObjectNav applications.

Previous works on 3DSG generation can be broadly categorized into offline methods and those that do not support open-set object detection, both of which present critical limitations. Several studies have made notable advancements in this domain, yet they remain constrained by specific drawbacks that hinder their applicability to ObjectNav. Hydra [1] demonstrates strong capabilities in real-time 3DSG generation; however, its lack of open-set object labeling restricts its flexibility, preventing agents from detecting and reasoning about previously unseen objects. HOV-SG [2] and ConceptGraph [3] incorporate vision-language features to enable open-set object representation, yet their methods require substantial computational resources, rendering them impractical for real-time applications. CLIO [4] integrates open-set object detection within a real-time 3DSG framework, but its reliance on 3D mesh reconstruction introduces unnecessary computational overhead, making it less efficient for rapid navigation. VOOM [5], an ellipsoid-based SLAM system, provides efficient object detection and optimization using elliptical representations; however, as a SLAM-based approach, it lacks the hierarchical and high-level semantic reasoning required for effective scene understanding and

navigation. While each of the methods contributes insights into 3DSG construction, their limitations, whether due to computational inefficiency, restricted open-set, or the absence of high-level semantic reasoning, highlight the need for a more versatile and efficient solution.

To address these challenges, we introduce a real-time 3DSG generation framework that supports open-set object detection and natural language-based navigation. Unlike prior methods that rely on closed-set object categories or computationally intensive scene reconstruction, our approach enables dynamic, incremental SG construction during environment exploration, allowing agents to process arbitrary user-specified object queries efficiently. At the core of our method is YOLO-World [6], an open-set object detector that provides bounding box predictions for a diverse range of objects, facilitating flexible SG updates. Inspired by VOOM [5], we incorporate an *ellipsoid-based representation* to model detected objects, offering a lightweight yet expressive alternative to dense point cloud or mesh-based reconstructions. This representation is further optimized to ensure robust and efficient object localization, significantly reducing computational overhead while preserving essential spatial and semantic information. Moreover, by integrating hierarchical 3DSG structures with large language models (LLMs), our system leverages context-aware reasoning, allowing agents to infer spatial relationships, understand abstract navigational cues, and adapt to dynamic environments in a manner akin to human navigation. Ultimately, our framework bridges the gap between real-time 3DSG construction, open-set object recognition, and natural language-driven navigation, offering a scalable way for ObjectNav in diverse and previously unseen environments.

To validate the effectiveness of the proposed framework for robotic navigation, we primarily evaluate on the Habitat-Matterport 3D (HM3D) dataset [7] under the challenging ObjectNav benchmark. ObjectNav requires an agent to navigate toward a target object specified by its semantic category, making it a task that directly links perception, mapping, and goal-directed planning. Across approximately 20 diverse indoor scenes, our method consistently outperforms baselines without stair-climbing in terms of *Success weighted by Path Length (SPL)*, demonstrating superior robustness and generalization in realistic navigation settings. In addition, we conduct complementary experiments on the ScanNet dataset [8], where we evaluate how effectively our object-based scene graph captures semantic and structural cues across seven representative indoor categories. In addition, we conduct complementary experiments on the ScanNet dataset [8], where we evaluate how effectively our object-based scene graph captures semantic and structural information. These findings validate the adaptability of our method to real-world environments, highlighting its potential for practical deployment in autonomous navigation tasks. The primary contributions of this paper are summarized as follows:

- To the best of our knowledge, this work introduces the first framework that integrates fast, open-set object detectors for object localization within 3D scene graph

(3DSG) construction, while employing an ellipsoid-based representation to structure the object graph. This approach facilitates real-time scene graph generation, enabling efficient and flexible representation of open-set objects.

- In ObjectNav tasks evaluated on realistic indoor scenes from the HM3D dataset, our method outperforms baselines with superior navigation efficiency and robustness.
- Experiments on the ScanNet dataset demonstrate that the proposed method can accurately capture 3D spatial information, validating its effectiveness in SG representation.

II. RELATED WORK

A. 2D Semantic Map

Recent research has explored the integration of semantic perception with robotic mapping to construct enriched environmental representations. Traditional approaches combine 2D semantic segmentation with spatial mapping, enabling robots to reason beyond purely geometric information. The advent of open-vocabulary detection models has further advanced this direction by improving adaptability and robustness in semantic mapping. VL-Map [9] projects embeddings from LSeg’s visual encoder [10] onto a 2D costmap, thereby supporting on-demand semantic queries through a text encoder. To enhance exploration capabilities, both VLFM [11] and OpenFMNav [12] incorporate frontier-based exploration strategies where frontiers are prioritized using vision-language models. While these methods demonstrate improved exploration efficiency, they remain limited by their reliance on 2D representations. The absence of explicit 3D spatial structures restrict their ability to accurately capture object positions in complex environments.

B. 3D Scene Graphs (3DSG)

Scene graphs, initially introduced by [13], are visually grounded data structures that represent a scene through objects, their attributes, and the pairwise relationships between them. While traditional scene graphs operate primarily in 2D image space, recent research has extended this concept to 3DSGs [14], [15], which incorporate spatial information to enable richer and more grounded representations of complex environments. ConceptGraph [3] combines SAM and CLIP to extract semantic information of objects and builds a minimum spanning tree (MST) based on 3D spatial relationships between objects. It further leverages LLMs to infer semantic relations between connected objects.

Hydra [1] introduces a hierarchical 3DSG structure that supports real-time processing, enabling the use of 3DSGs in a wider range of applications. CLIO [4] extends Hydra’s framework by integrating open-vocabulary object detectors, making the resulting scene graph more flexible and applicable to diverse downstream tasks. Building upon this, HOV-SG [2] utilizes the hierarchical property of the 3DSG to enable top-down reasoning with LLMs, facilitating efficient object search via traditional tree traversal methods. SayNav [16] takes this further by applying scene graphs to multi-object

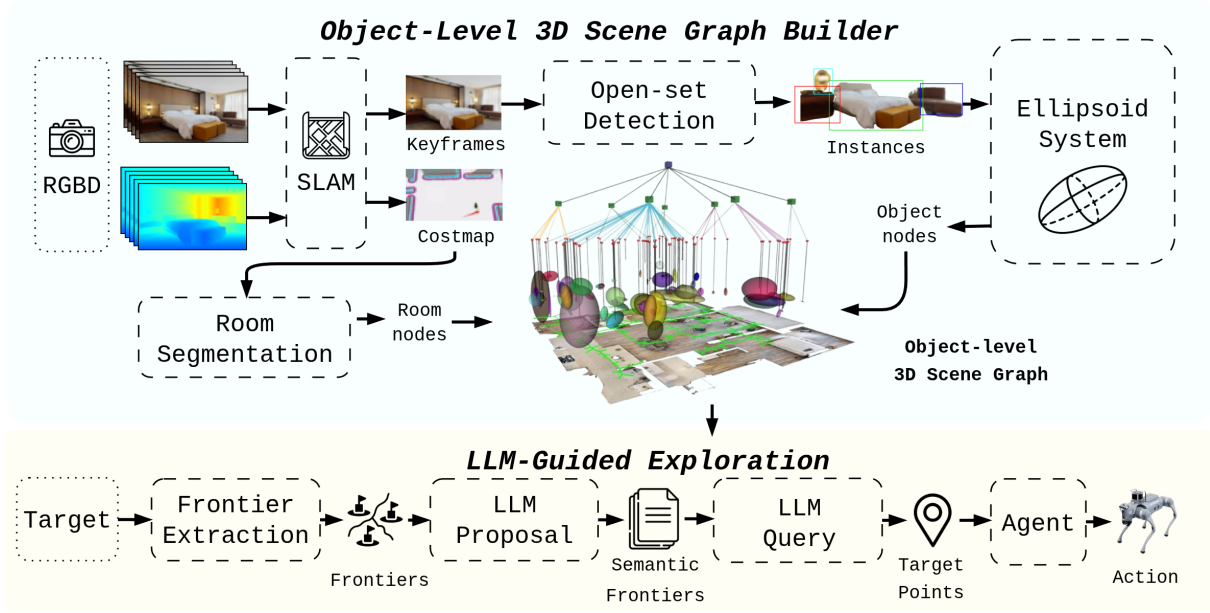


Fig. 2. Architecture of our framework.

navigation tasks. It employs LLMs and 3DSGs for short-term planning but lacks the ability to identify unexplored areas in the environment. SG-Nav [17] addresses this gap by incorporating the concept of frontiers and feeding subgraphs near these frontiers into the LLM, thereby improving exploration efficiency. UniGoal [18] further enhances scene graph-based navigation by introducing a graph-matching mechanism that enables 3DSGs to be used effectively in navigation tasks.

While the above works primarily focus on the spatial and structural aspects of 3D scene graphs for reasoning and navigation, another line of research [19], [20] emphasizes 3D semantic scene graphs. Methods such as FROSS [21] model objects as 3D Gaussian distributions and focus on capturing semantic relationships. Since our work aims to construct a hierarchical spatial scene graph that encodes geometric organization rather than semantic associations, these methods are not directly comparable to ours and are therefore beyond the scope of this paper.

III. METHODOLOGY

A. Problem Definition

This work tackles the challenge of real-time object navigation and hierarchical 3D scene graph (3DSG) construction. The agent’s goal is to explore an unknown indoor environment, locate a target object, and simultaneously build a hierarchical 3DSG that represents the scene. A key challenge is balancing efficient exploration with path planning. To address this, we propose a framework that constructs a hierarchical 3DSG in real-time during exploration. The agent continuously detects objects and incrementally builds the scene graph. This structured representation enhances the agent’s understanding of the environment’s layout. In

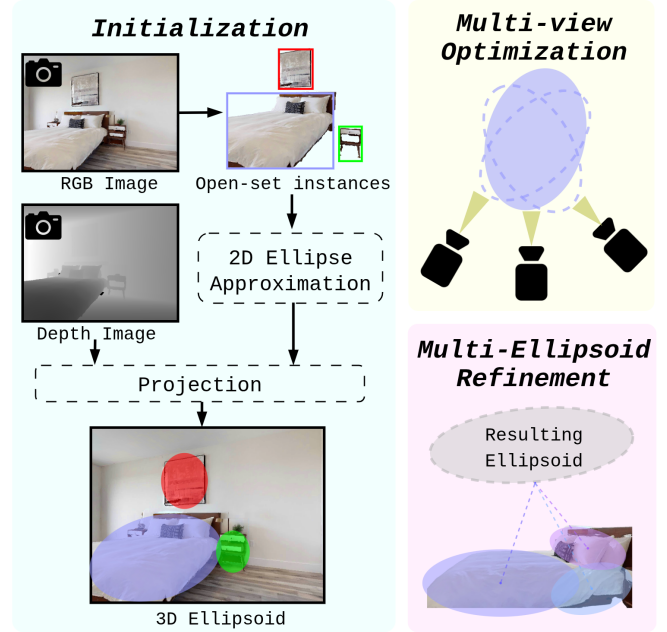


Fig. 3. Overview of our ellipsoid construction pipeline. Given RGB-D inputs and detected object instances, we initialize with 2D ellipse approximations and projected to 3D space, followed by multi-view optimization and multi-ellipsoid refinement to produce consistent and accurate 3D ellipsoids.

addition, leveraging a large language model (LLM) allows the agent to infer high-priority areas for exploration, leading to more efficient navigation toward the target.

B. Framework Overview

Our scene graph is composed of two main components: scene graph builder, and exploration module, which is shown

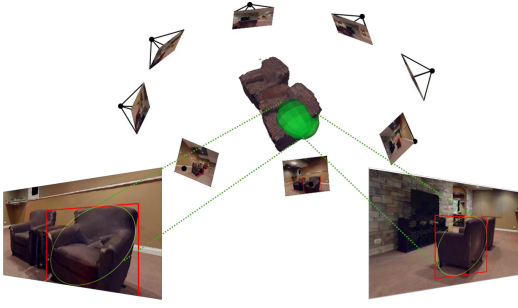


Fig. 4. Demonstration of the ellipsoid optimization. Initially, the original ellipsoid is projected onto each frame, and the 2nd-order Wasserstein distance is used to compute the error between the projected ellipses and the ellipse stored in each frame. By minimizing the total error, an improved ellipsoid can be obtained.

in Fig. 5. First, we input the RGB image into the detection model, and using the agent’s position with depth information project the detected objects back into 3D coordinates. Through room segmentation, we divide the explored areas into smaller regions based on spatial relationships. Using the Voronoi diagram-based navigation, we can plan paths to any location while maintaining the maximum distance from obstacles, which minimizes the risk of collisions during the agent’s movement. Finally, frontier exploration integrates all the information above to generate a complete scene graph. Along with the target object, this scene graph is then used to query an LLM, which, with its understanding of the environment layout, helps select the most suitable area for further exploration. This integrated approach optimizes exploration and navigation within the unknown environment.

C. 3D Object Construction

Our object construction method is adapted from VOOM [5]. The following sections will go through the process of obtaining the object ellipsoid step-by-step.

1) *Object Initialization*: When a new RGB-D image is obtained, we first perform the robot localization. Unlike VOOM, which utilizes ORB-SLAM2 [22] as its SLAM system, we choose RTAB-Map [23] as our SLAM system. This decision is based on the need to obtain the current environment’s costmap during the construction of the 3D Scene Graph. Using RTAB-Map, we can seamlessly integrate these processes.

Subsequently, instance segmentation is performed on each keyframe from RTAB-Map to initialize the ellipsoid of 3D objects. Although YOLO [24] series models can quickly generate masks, their closed-set nature limits their applicability in diverse scenarios. Therefore, we opted for an open-set detector. In approaches such as HOV-SG [2] and ConceptGraph [3], segmentation is typically achieved by applying SAM [25] first, followed by CLIP [26] to obtain semantic information for each mask. However, SAM requires significant computational and time resources to process an entire image, which prevents these pipelines from achieving real-time performance. To address this limitation, we employ YOLO-World [6], an open-set object detector, to first identify

2D bounding boxes of objects. Subsequently, SAM is used to perform segmentation, reducing the amount of computational resources. This optimization enables the entire pipeline to achieve real-time performance.

By utilizing the camera pose estimation from RTAB-Map and the object masks generated by YOLO-World/SAM, we can efficiently initialize the object’s ellipsoid. To reduce the impact of noise on the depth map, the depth is estimated by randomly sampling a few points within the region and taking the average as the final estimation.

2) *Object Association*: To prevent duplicate object creation, it is necessary to verify whether an object has already been instantiated. In VOOM, all constructed objects are projected onto the keyframe to generate a set of 2D ellipses and then evaluated using the normalized Wasserstein distance.

However, we found that this method struggled with large objects, such as sofas or tables, where the entire object cannot be fully captured within a single frame. In such cases, it becomes challenging to associate them correctly. Therefore, for objects that project onto the keyframe and exceed the boundaries, we employ the following formula for association:

$$\text{Intersection} = \max \left(\frac{\text{Intersection Area}}{\text{Ellipse}_1 \text{ Area}}, \frac{\text{Intersection Area}}{\text{Ellipse}_2 \text{ Area}} \right) \quad (1)$$

To simplify the calculation of the ellipse area and the intersection area, we approximate them using the volume of the bounding box enclosing the ellipse. Additionally, in scenarios where multiple objects of the same category are projected onto similar locations, VOOM selects the one closest to the camera pose, which may lead to incorrect association. In contrast, our method selects the object whose estimated depth most closely matches the expected depth of the target depth, resulting in more accurate object association.

3) *Ellipsoid Optimization*: If an object is successfully associated, the segmented object mask from the keyframe and the corresponding camera pose are recorded for that object. Once data from more than three frames is recorded, object optimization is performed, which is shown in Fig. 4.

Before optimization, RTAB-Map will update the latest pose for each keyframe. Following VOOM’s [5] approach, the 2nd-order Wasserstein distance is used to measure the error between the estimated ellipse and the observed ellipse. Finally, g2o [27] is employed to minimize the error across all keyframes, resulting in a more accurate estimate of the ellipsoid.

4) *Ellipsoid Refinement*: Although we have improved the association process for large objects compared to VOOM, there are still cases where association fails or objects cannot be accurately represented by a single ellipsoid, such as L-shaped sofas. To address these issues, we introduced a post-processing step.

During object association, all current object ellipsoids are projected onto the keyframe. If multiple ellipsoids of the same class appear on the same keyframe, we calculate the 3D spatial distance between these ellipsoids. If the distance is

TABLE I

COMPARISON BETWEEN DIFFERENT METHODS ON THE HM3D OBJECTNAV BENCHMARK. OUR METHOD OUTPERFORMS MOST OF THE BASELINES IN BOTH SUCCESS RATE (SR) AND SUCCESS WEIGHTED BY PATH LENGTH (SPL).

Method	SR (%)	SPL
L3MVN	50.4	0.231
OpenFMNav	52.5	0.244
SG-Nav	54.0	0.249
UniGoal	54.5	0.251
Object-Level SG (Ours)	54.2	0.266

below a certain threshold, they are linked together. In the ellipsoid optimization stage, each linked ellipsoid is optimized independently. However, when representing the object, the centers of the linked ellipsoids are used to calculate the true center point of the object. This approach ensures a more accurate representation of complex or large objects.

D. Frontier Exploration

We adopt frontier exploration as the basis for navigating unknown environments, where a frontier is defined as the boundary between explored and unexplored regions of the map. To improve the efficiency and robustness of frontier selection, we extend this classical approach with semantic-aware reasoning, allowing the agent to prioritize frontiers that are not only spatially informative but also contextually relevant to the searching goal.

1) *Semantic Frontier Proposal*: Each frontier is characterized by a local semantic subgraph extracted from the global scene graph, in which nodes represent objects nearby the frontier. Incorporating semantic information enables the agent to distinguish frontiers that lead to distinct areas. To avoid redundant targets that lie in nearly the same location, we compute pairwise Euclidean distances between all frontiers and discard any frontier whose distance falls below a defined threshold.

2) *LLM-Guided frontier exploration*: Based on the room segmentation of graph, we assign each room a coarse room label by prompting the LLM with current scene graph, thereby focuses attention on semantically coherent areas. After this room annotation, we invoke a LLM with the full scene graph, brief descriptors of each frontier subgraph, and the target in a single prompt. By leveraging its internal reasoning, the top N frontiers most likely to lead to the target object are collected into the candidate set $\mathcal{F}$, with each frontier $i \in \mathcal{F}$ assigned a relevance probability P_i . However, relying solely on these probabilities can lead the agent to pursue highly relevant but impractically distant frontiers, resulting in wasted travel and longer search times. To address this, we compute the true navigational cost c_i to each frontier by querying our path planner, and combine this cost with P_i for each frontier. We select the best frontier $f_b \in \mathcal{F}$ by:

$$f_b = \arg \max_{i \in \mathcal{F}} \frac{P_i}{d_i}$$

TABLE II

ABLATION STUDY, COMPARISON BETWEEN WITH OR WITHOUT LLM GUIDED

Method	SR (%)	SPL
Without LLM	22.0	0.113
Nemotron-9B	52.2	0.248
Llama3-70B	54.2	0.266

IV. EXPERIMENTAL RESULTS

A. Object Navigation

This section shows the experimental setup and metrics used to evaluate our method, and benchmarking baselines to demonstrate improvements in handling language-guided navigation tasks. We aim to validate the navigation effectiveness of our framework, through our experiment.

1) *Environmental Setups*: We use the Habitat simulator [28] to evaluate the performance on the Object Navigation task. Our evaluation leverages the Habitat-Matterport 3D dataset [7], which includes an extensive set of real-world indoor scenes with a wide variety of room layouts and object configurations, which enables a complete assessment of our method’s generalization capabilities. For our experiments, we utilize the validation set of the HM3D dataset, which includes 20 diverse indoor scenes, each containing 6 distinct object configurations for the Object Navigation task. From this set, we randomly selected several scenes with varying layouts to evaluate performance. In the simulation environment, the robot navigates using discrete actions: {move_forward, turn_left, turn_right, look_up, look_down, finish}. Each forward movement covers a distance of 0.25 meters, while rotations are performed in 30-degree increments per step. The robot’s radius is set to 0.1 meters, and the success distance is set to 0.13 meters. Additionally, RGB-D images and semantic information are extracted from the simulation environment at each step. The robot is required to find the target object within a limit of 500 steps.

2) *Evaluation metrics*: For evaluation, we adopt the Success Rate (SR), which measures the agent’s ability to find the target objects, as commonly used in prior works. Additionally, we report the Success weighted by Path Length (SPL) to assess the efficiency of the agent in finding the objects relative to the path taken. SPL rewards the agent for achieving success while minimizing the path length, offering a measure of navigation efficiency.

3) *Baselines*: We benchmark our method against several previous works, focusing on open-set and zero-shot object navigation. The experimental settings for the baselines are taken from [12].

- ZSON [29]: CLIP-based zero-shot object navigation.
- GoW [30]: Utilizing GLIP for object detection and explore by frontier exploration.
- ESC [31]: GLIP-based 2D mapping method, and combine with the LLM for planning.
- L3MVN [32]: 2D semantic map combining with LLM guided frontier exploration.



Fig. 5. Qualitative results of object navigation.

- OpenFMNav [12]: 2D semantic map and navigating with LLM-guided frontier exploration.
- SG-Nav [17]: 3D scene graph with LLM-based zero-shot frontier exploration.
- UniGoal [18]: Zero-shot navigation based on goal matching.

4) *Result*: The main results are presented in Table I. Our method outperforms most of the baselines in both Success Rate (SR) and Success weighted by Path Length (SPL). Specifically, our method surpasses the ZSON approach, which is trained on the Image Navigation task and transferred to Object Navigation. This demonstrates the critical role of a semantic-rich map when navigating to a specified object.

Furthermore, our method outperforms ESC, which projects 3D information into a 2D semantic map as prior knowledge for LLM-guided frontier exploration. While ESC combines LLM common sense with frontier exploration, it still lacks the full benefits of 3D spatial awareness. In contrast, our method employs a 3D scene graph to encode environmental information into a tree-like structure, retaining much more semantic detail, which is crucial for the effectiveness of object navigation.

5) *Ablation Study*: To assess the impact of LLM-guided exploration, we conducted an ablation study. This involved removing the LLM from the frontier selection process and reverting to a random selection baseline, thereby eliminating the influence of common-sense guidance. Furthermore, to evaluate the generality and robustness of our pipeline, we also conducted experiments using nemotron-nano-9B [33] to test the framework’s adaptability to different LLMs.

The results, summarized in Table II, demonstrate that while our pipeline leverages a hierarchical and semantic 3D scene graph, the incorporation of LLM-guided planning is crucial for achieving superior performance. This finding underscores that LLM-guided frontier exploration is not merely an addition but a key component, significantly enhancing

TABLE III
SCENE GRAPH EVALUATION WITH R @ 1.5

Method	R @ 1.0	R @ 1.5
CLIo [4]	0.17	0.22
Ours	0.23	0.26

navigation efficiency in finding target objects.

Furthermore, we validated the robustness and generality of our framework. We observed that even with a smaller 9B model, the system still performed well, successfully guiding the exploration task. This demonstrates that our method is not dependent on a single, large-scale model. Instead, it indicates the core framework is highly adaptable and can be generalized to various LLMs, including more lightweight versions that are better suited for practical deployment on resource-constrained robotic systems.

B. Scene Graph Evaluation

We assess the accuracy of our proposed method for reconstructing object-based scene graphs. This subsection details the experimental setup and evaluation metrics used to validate our approach. Owing to the variability in scene graph construction across different structure definitions, our analysis is confined to the primary object layer.

1) *Environmental Setups*: Our evaluation is performed on the ScanNet [8] dataset, which comprises a diverse collection of indoor scenes from the real world featuring various types of room, demonstrating the generalization of our method. To ensure a comprehensive assessment, experiments are conducted across seven commonly encountered categories, with three scenes randomly selected per category. All experiments adhere to the simulation settings described in section IV-A.1.

2) *Evaluation metrics*: In addition to conventional recall metrics, we propose the R @ d evaluation metric to directly assess the accuracy of the 3D positions estimated by our

TABLE IV
RUNTIME EVALUATION RESULT

Method	Frame per Second (FPS)
CLIo [4]	3.56
Ours	9.86

method. $R @ d$ quantifies the percentage of predicted object positions that fall within a predefined Euclidean distance threshold d from their corresponding ground truth positions.

3) *Result*: The primary results are presented in Table III. Since both our approach and the baseline are open-set methods, the framework can be deployed across diverse environments. The results indicate that the ellipsoid-based system, with properly optimized and merged systems, outperforms the baseline. This shows that our framework captures more accurate spatial information, which in turn enhances the navigation system.

C. Runtime Evaluation

To assess the runtime performance of our framework, an evaluation was conducted and compared with a baseline method that also implements open-set, real-time scene graph construction. Due to the reliance on the RTAB-Map skip-frame method to reduce redundant computations, direct evaluation of the real-time performance of our framework with RTAB-Map is not feasible. Instead, images and odometry poses are obtained directly from the simulator during experiments, enabling an accurate assessment of the performance of our method and highlighting the efficiency of our object-based framework.

1) *Environmental Setups*: Runtime performance is evaluated using an NVIDIA RTX 3060 12G GPU and an Intel i5-13500 CPU. We used frames per second (FPS) as the primary metric to quantify the efficiency of our method relative to the baseline.

2) *Result*: The results presented in Table IV demonstrate that our method outperforms the baseline, successfully constructing object-level graph in real time. Emphasizing the efficiency of our object-based representation in accurately capturing spatial relationships while maintaining computational efficiency.

(J:

3) *LLM*: LLM used Llama-3? via self-hosted NVIDIA NIM on eight L40 GPUs.)

V. CONCLUSION

Our framework employs a fast, open-set object detector to construct an object-level 3D scene graph, dynamically capturing and organizing environmental features into a hierarchical representation. It provides spatial relationships and semantic information in 3D space, essential for guiding the robot. Additionally, we integrate LLMs to guide frontier exploration based with their common-sense reasoning. The LLM leverages this object-based 3D scene graph to prioritize areas likely to contain target objects, enhancing the agent's ability to efficiently search for certain object.

Through our experiments, we demonstrate that both the object-level 3D scene graph and LLM-guided frontier exploration are pivotal in improving object navigation performance, validated on the HM3D ObjectNav benchmark.

A primary limitation of the proposed framework emerges in environments with multi-level structural, where the agent and target object reside on separate levels. In such scenarios, our navigation module is restricted in single floor and unable to traverse staircases to move between floors, leading to failures.

REFERENCES

- [1] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," *Robotics: Science and Systems*, 2022. 1, 2
- [2] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3d scene graphs for language-grounded robot navigation," *Robotics: Science and Systems*, 2024. 1, 2, 4
- [3] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. M. de Melo, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, "Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning," 2023. [Online]. Available: <https://arxiv.org/abs/2309.16650> 1, 2, 4
- [4] D. Maggio, Y. Chang, N. Hughes, M. Trang, D. Griffith, C. Dougherty, E. Cristofalo, L. Schmid, and L. Carlone, "Clio: Real-time task-driven open-set 3d scene graphs," 2024. [Online]. Available: <https://arxiv.org/abs/2404.13696> 1, 2, 6, 7
- [5] Y. Wang, C. Jiang, and X. Chen, "Voom: Robust visual object odometry and mapping using hierarchical landmarks," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 10298–10304. 1, 2, 4
- [6] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, "Yolo-world: Real-time open-vocabulary object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16901–16911. 2, 4
- [7] S. K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. M. Turner, E. Undersander, W. Galuba, A. Westbury, A. X. Chang, M. Savva, Y. Zhao, and D. Batra, "Habitat-matterport 3d dataset (HM3d): 1000 large-scale 3d environments for embodied AI," in *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021. [Online]. Available: <https://arxiv.org/abs/2109.08238> 2, 5
- [8] A. Dai, A. X. Chang, M. Savva, M. Halber, T. Funkhouser, and M. Nießner, "ScanNet: Richly-annotated 3d reconstructions of indoor scenes," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 5828–5839. 2, 6
- [9] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 10608–10615. 2
- [10] B. Li, K. Q. Weinberger, S. Belongie, V. Koltun, and R. Ranftl, "Language-driven semantic segmentation," 2022. [Online]. Available: <https://arxiv.org/abs/2201.03546> 2
- [11] N. Yokoyama, S. Ha, D. Batra, J. Wang, and B. Bucher, "Vlfm: Vision-language frontier maps for zero-shot semantic navigation," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 42–48. 2
- [12] Y. Kuang, H. Lin, and M. Jiang, "Openfmnav: Towards open-set zero-shot object navigation via vision-language foundation models," in *Findings of the Association for Computational Linguistics: NAACL* 2024, 2024, pp. 338–351. 2, 5, 6
- [13] J. Johnson, R. Krishna, M. Stark, L.-J. Li, D. A. Shamma, M. S. Bernstein, and L. Fei-Fei, "Image retrieval using scene graphs," in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 3668–3678. 2
- [14] I. Armeni, Z.-Y. He, J. Gwak, A. R. Zamir, M. Fischer, J. Malik, and S. Savarese, "3d scene graph: A structure for unified semantics, 3d space, and camera," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 5664–5673. 2

- [15] A. Rosinol, A. Gupta, M. Abate, J. Shi, and L. Carlone, “3d dynamic scene graphs: Actionable spatial perception with places, objects, and humans,” *arXiv preprint arXiv:2002.06289*, 2020. 2
- [16] A. Rajvanshi, K. Sikka, X. Lin, B. Lee, H.-P. Chiu, and A. Velasquez, “Saynav: Grounding large language models for dynamic planning to navigation in new environments,” in *Proceedings of the International Conference on Automated Planning and Scheduling*, vol. 34, 2024, pp. 464–474. 2
- [17] H. Yin, X. Xu, Z. Wu, J. Zhou, and J. Lu, “Sg-nav: Online 3d scene graph prompting for llm-based zero-shot object navigation,” *Advances in neural information processing systems*, vol. 37, pp. 5285–5307, 2024. 3, 6
- [18] H. Yin, X. Xu, L. Zhao, Z. Wang, J. Zhou, and J. Lu, “Unigoal: Towards universal zero-shot goal-oriented navigation,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 19 057–19 066. 3, 6
- [19] J. Wald, H. Dharm, N. Navab, and F. Tombari, “Learning 3d semantic scene graphs from 3d indoor reconstructions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020. 3
- [20] S.-C. Wu, K. Tateno, N. Navab, and F. Tombari, “Incremental 3d semantic scene graph prediction from rgb sequences,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2023, pp. 5064–5074. 3
- [21] H.-Y. Hou, C.-Y. Lee, M. Sonogashira, and Y. Kawanishi, “Fross: Faster-than-real-time online 3d semantic scene graph generation from rgb-d images,” *arXiv preprint arXiv:2507.19993*, 2025. 3
- [22] R. Mur-Artal and J. D. Tardós, “Orb-slam2: An open-source slam system for monocular, stereo, and rgb-d cameras,” *IEEE transactions on robotics*, vol. 33, no. 5, pp. 1255–1262, 2017. 4
- [23] M. Labbé and F. Michaud, “Rtab-map as an open-source lidar and visual simultaneous localization and mapping library for large-scale and long-term online operation,” *Journal of Field Robotics*, vol. 36, no. 2, pp. 416–446, 2019. 4
- [24] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, “Yolov9: Learning what you want to learn using programmable gradient information,” *arXiv preprint arXiv:2402.13616*, 2024. 4
- [25] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo et al., “Segment anything,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4015–4026. 4
- [26] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark et al., “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PMLR, 2021, pp. 8748–8763. 4
- [27] R. Kümmerle, G. Grisetti, H. Strasdat, K. Konolige, and W. Burgard, “g 2 o: A general framework for graph optimization,” in *2011 IEEE international conference on robotics and automation*. IEEE, 2011, pp. 3607–3613. 4
- [28] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Parikh, and D. Batra, “Habitat: A Platform for Embodied AI Research,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019. 5
- [29] A. Majumdar, G. Aggarwal, B. Devnani, J. Hoffman, and D. Batra, “Zson: Zero-shot object-goal navigation using multimodal goal embeddings,” 2023. [Online]. Available: <https://arxiv.org/abs/2206.12403> 5
- [30] S. Y. Gadre, M. Wortsman, G. Ilharco, L. Schmidt, and S. Song, “Cows on pasture: Baselines and benchmarks for language-driven zero-shot object navigation,” 2022. [Online]. Available: <https://arxiv.org/abs/2203.10421> 5
- [31] K. Zhou, K. Zheng, C. Pryor, Y. Shen, H. Jin, L. Getoor, and X. E. Wang, “Esc: Exploration with soft commonsense constraints for zero-shot object navigation,” 2023. [Online]. Available: <https://arxiv.org/abs/2301.13166> 5
- [32] B. Yu, H. Kasaei, and M. Cao, “L3mvn: Leveraging large language models for visual target navigation,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, Oct. 2023, p. 3554–3560. [Online]. Available: <http://dx.doi.org/10.1109/IROS55552.2023.10342512> 5
- [33] A. Basant, A. Khairnar, A. Paithankar, A. Khattar, A. Renduchintala, A. Malte, A. Bercovich, A. Hazare, A. Rico, A. Ficek et al., “Nvidia nemotron nano 2: An accurate and efficient hybrid mamba-transformer reasoning model,” *arXiv preprint arXiv:2508.14444*, 2025. 6